

LLMs for research: quality, speed and cost

Comparison with Pi, replaying project tickets from Imperial Dragon Harness workflows

Ten tickets randomly sampled from the author's project history.

Local and hosted models replay the same tasks; three judges score the deliverables out of 30.

We compare quality, elapsed time and cost, including incidents and retries.

Luna 6 medium: the best observed trade-off

For both Claude and ChatGPT, moving up a model tier while reducing effort offers no clear advantage here: Opus 5.5 low versus Sonnet 5.5 medium, and Sol 6.1 low versus Luna 6 medium, have similar mean quality at higher cost. Sonnet is also faster than Opus; Luna has comparable speed to Sol. Haiku 5.5 matches Sonnet on quality (21.7 versus 21.4/30) for 0.13 *instead of* 0.30, but is three times slower.

Limitation: one model fills all three roles, which may disadvantage a model built for one.

Qwen 3.8 Flash-Next xhigh: a local option when quality comes first

With Strata and IQ3_S quantization, this setup has the highest mean score: 26.2/30, taking an average of 46 minutes per ticket. It is better suited to asynchronous work. Its lead is sample-sensitive: excluding ticket 0333 brings its mean down to Mistral's.

The local Qwen 3.8 setup matches hosted reference quality

The best local Qwen 3.8 setup matches the quality of commercial reference setups and improves substantially on Qwen 3.6. The gain combines generation, quantization, runtime and effort. Qwen runs at xhigh versus medium for the references: more verbose output and longer elapsed time.

Latency and shared-GPU serialization limit local interactive use

On the test workstation, latency and the need to serialize tasks on shared GPUs severely limit interactive use, while APIs allow highly parallel workflows. Future local models potentially derived from Qwen 4 and Mistral 4, hardware upgrades and daily runtime improvements may ease these constraints.

Mistral Large 4: promising quality, reliability still costly

Mistral Large 4, announced on October 6, 2026, shows promising quality. However, hosting incidents and loops observed in its integration with Pi still hurt reliability, speed and cost.

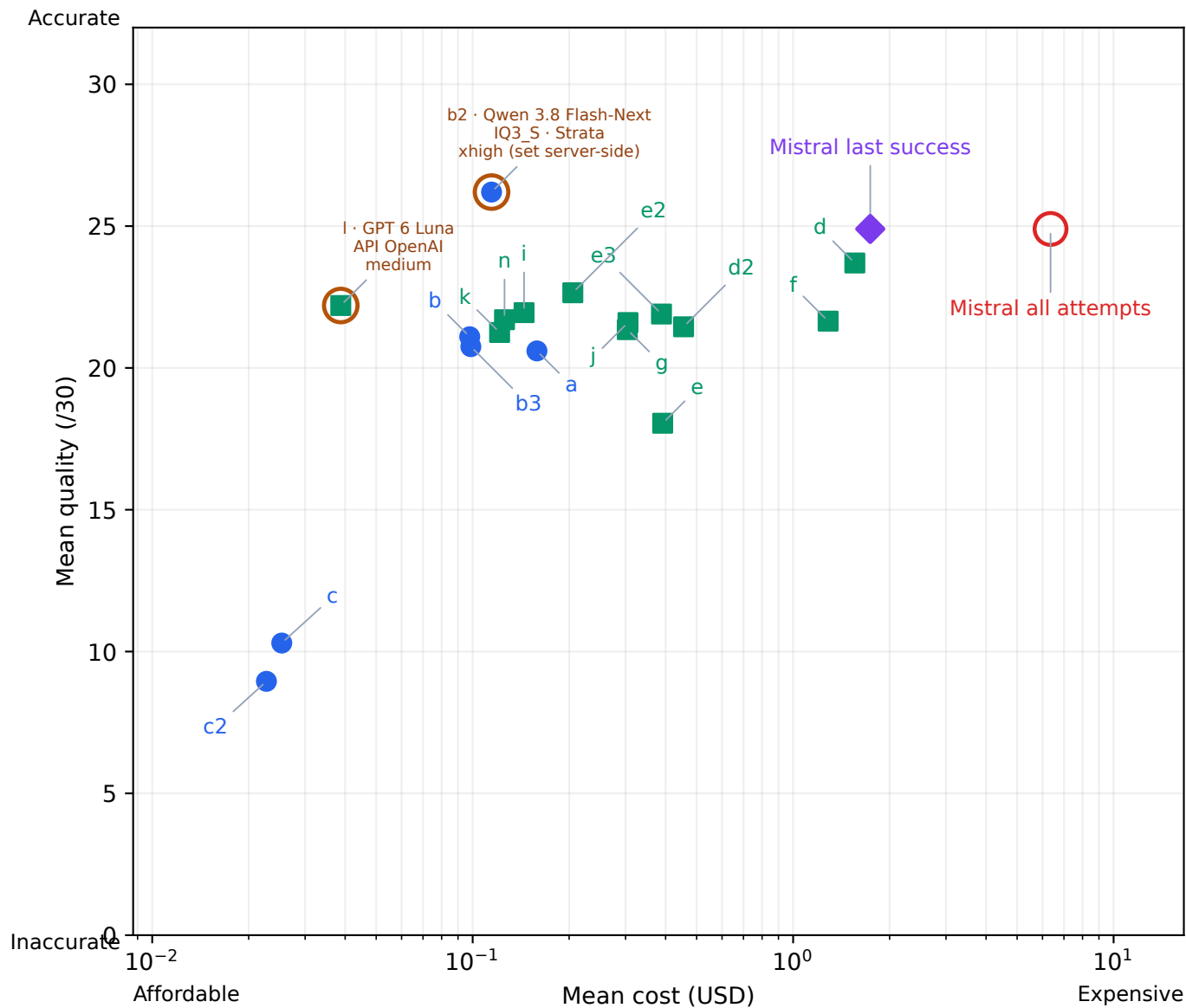
Results by model

Model · variant / engine · effort	Attempted	Completed / 10	Mean quality /30	Mean time (min / ticket)	Mean cost (USD / ticket)	Mean generated tokens / ticket
a · Qwen 3.8 27B Q4_K_M · llama.cpp · xhigh (set server-side)	16	9 / 10	20.60	63.87	0.16	68 016
b · Qwen 3.8 Flash-Next Coder IQ1_M · Strata · pruned · xhigh (set server-side)	14	10 / 10	21.10	39.41	0.10	64 702
b2 · Qwen 3.8 Flash-Next IQ3_S · Strata · xhigh (set server-side)	11	10 / 10	26.20	46.11	0.11	71 072
b3 · Qwen 3.8 Flash-Next Q2_0 · Strata · xhigh (set server-side)	10	9 / 10	20.75	39.73	0.10	71 416
c · Qwen 3.6 35B-A3B UD-Q4_K_XL · llama.cpp · effort not measured	10	10 / 10	10.30	10.21	0.03	24 584
c2 · Qwen 3 Coder 30B-A3B UD-Q4_K_XL · llama.cpp · effort not measured	10	9 / 10	8.95	9.15	0.02	11 477
d · Claude Opus 5.5 API Anthropic · medium	12	10 / 10	23.70	9.07	1.56	24 773
d2 · Claude Opus 5.5 API Anthropic · low	10	10 / 10	21.45	3.03	0.46	8 666
e · GPT 6 Sol API OpenAI · medium	10	10 / 10	18.05	4.91	0.39	7 548
e2 · GPT 6.1 Sol API OpenAI · low	10	10 / 10	22.65	5.15	0.21	4 687
e3 · GPT 6.1 Sol API OpenAI · medium	10	10 / 10	21.90	8.38	0.39	9 425
f · GLM 5.3 API Mistral · high	10	10 / 10	21.65	10.58	1.29	46 748
g · Claude Sonnet 5.5 API Anthropic · medium	10	10 / 10	21.35	2.09	0.30	8 896
i · DeepSeek Flash API DeepSeek · high	10	10 / 10	21.95	9.58	0.14	63 848
j · GLM 5.3 Flash API OpenRouter · high	15	9 / 10	21.60	26.96	0.31	46 654
k · Mimo v2.6 Flash API OpenRouter · medium	13	9 / 10	21.25	60.06	0.12	125 159
l · GPT 6 Luna API OpenAI · medium	10	10 / 10	22.20	5.02	0.04	16 052
mL · Mistral Large 4 · Last success API Mistral + OR (0874) · off	35	10 / 10	24.90	30.60	1.74	59 605
mA · Mistral Large 4 · All attempts API Mistral + OR (0874) · off	35	10 / 10	24.90	96.32	6.36	196 233
n · Claude Haiku 5.5 API Anthropic · medium	10	10 / 10	21.70	6.11	0.13	26 511

Means per ticket, including failures; incomplete series cover evaluated tickets only. Local cost: electricity only. Mistral last success: ten successes, including 0874 via OpenRouter. Mistral all attempts: the same scores, with costs and times from 25 retained direct attempts plus the OR replacement for 0874; eight other OR attempts and one direct configuration incident are excluded. “Completed” counts delivered tickets, not attempts. The 9/10 rows reflect four two-hour timeouts (a and b3: 0874; j: 0333; k: 0211) and one termination without a valid deliverable (c2: 0470). These failures retain a zero score and consumed resources rather than being replaced by a successful retry; Mistral’s different treatment is explained in the limitations.

Quality / cost

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures



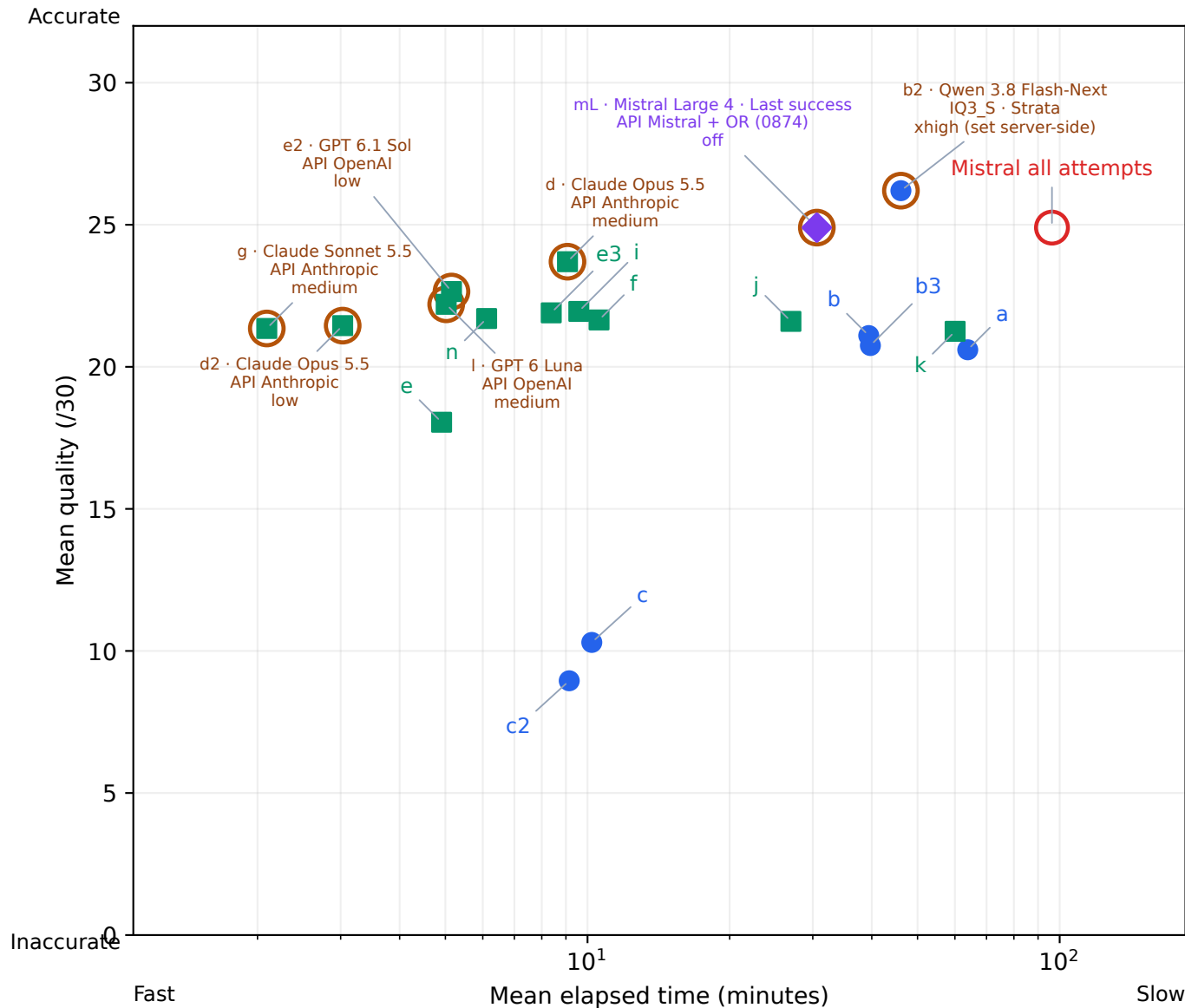
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API Anthropic · medium

Blue: local; green: hosted. Mistral: purple diamond = last success, red circle = all attempts. Cost and time use logarithmic scales. Orange ring: Pareto frontier (complete series, mean quality ≥ 15/30). Local cost = electricity only; hosted cost = API.

Quality / speed

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures



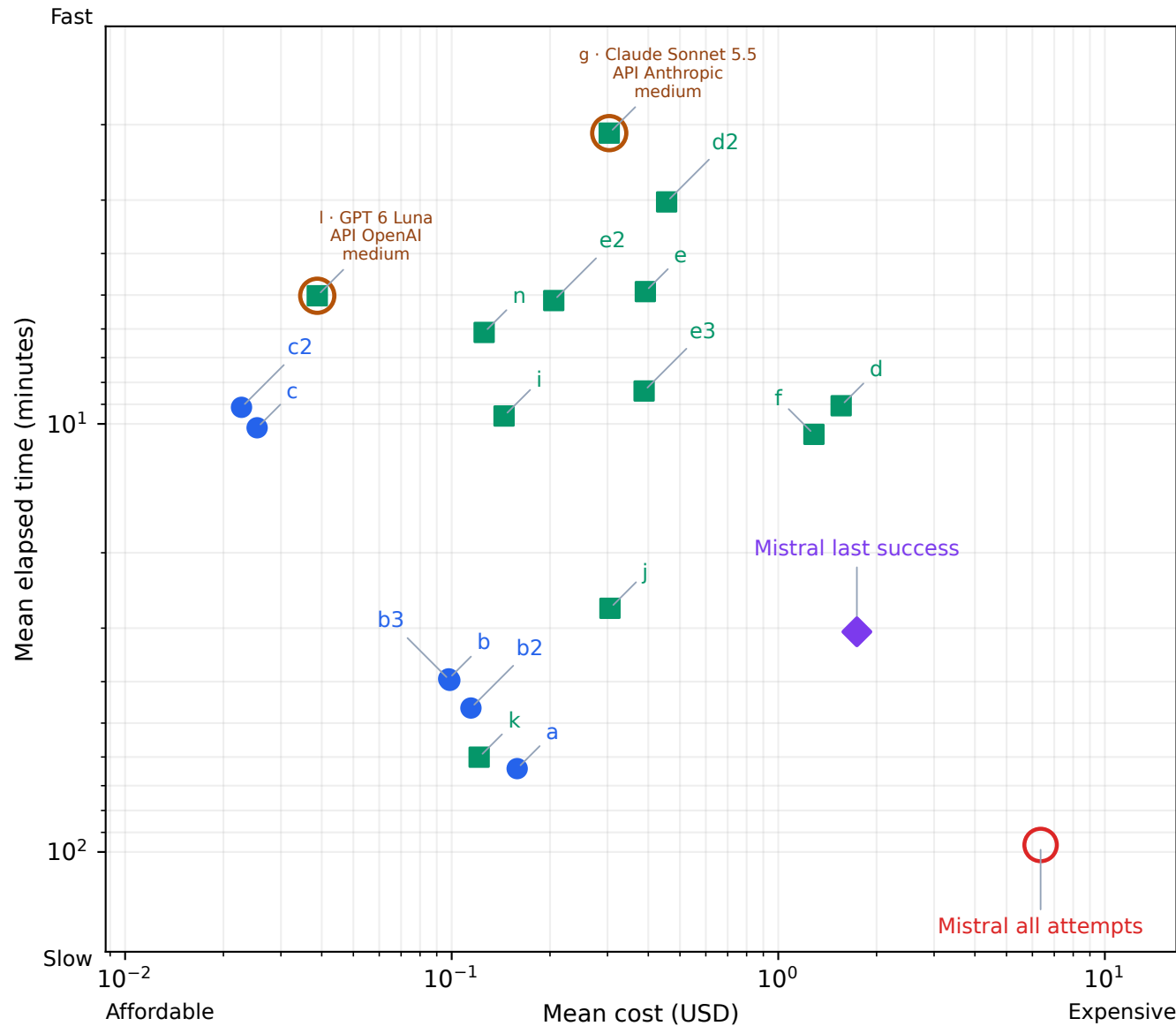
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Speed / cost

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures



Model · variant / engine · effort

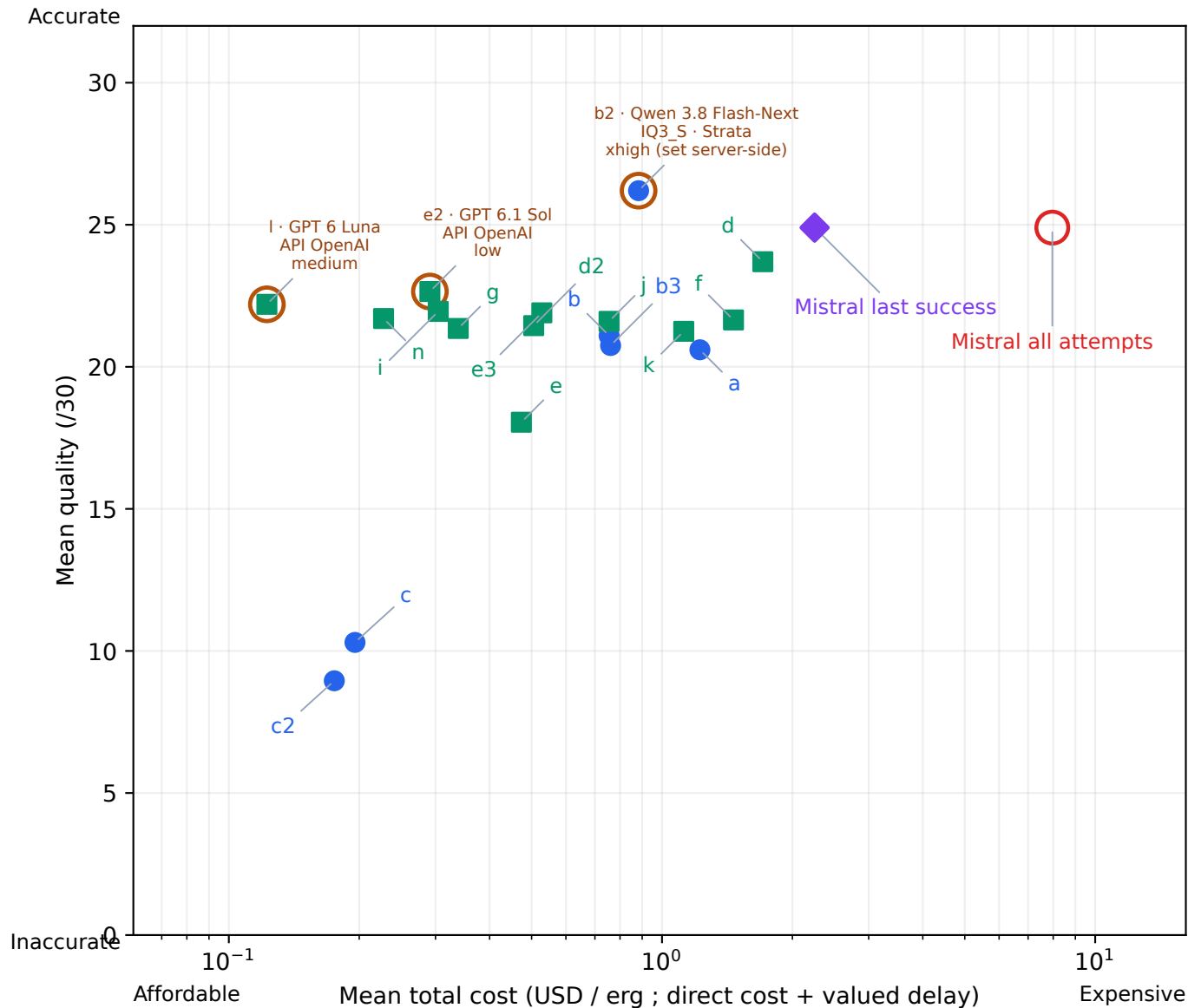
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Quality / total cost — delay valued at 1 USD/(erg·h)

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures

Per average erg: total cost = direct cost + 1 × time / 3,600; then averaged. Time in seconds, DNF included.



Model · variant / engine · effort

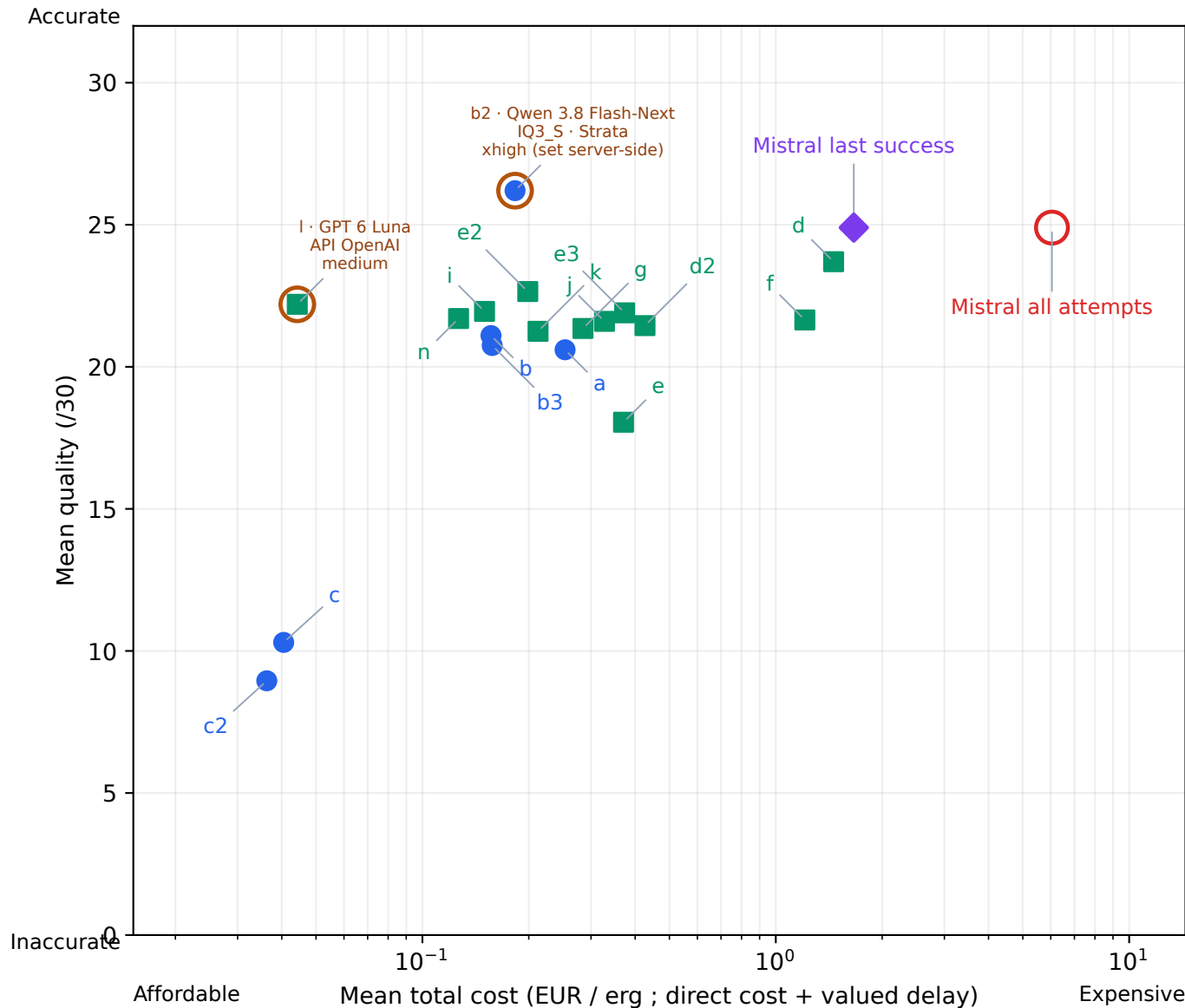
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Quality / total cost — delay valued at 0.10 EUR/(erg·h)

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures

Per average erg: direct cost in USD / 1.08 + 0.10 × time / 3,600 (seconds), then averaged. DNF included.



Model · variant / engine · effort

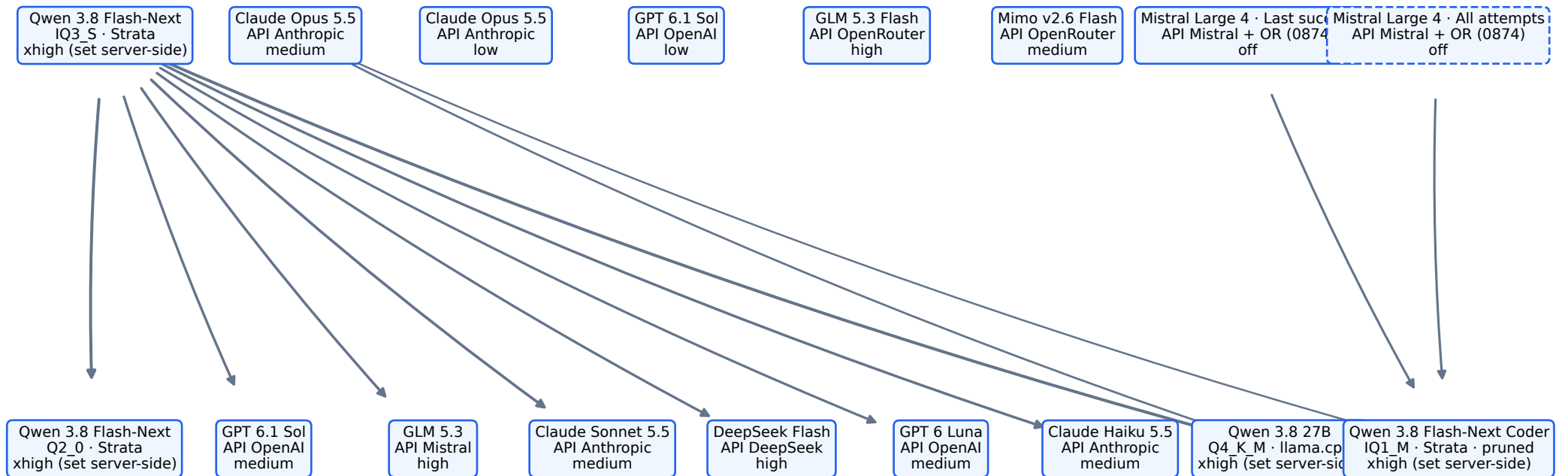
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Source: minh.ha-duong@cncs.fr, 2026-10-08

Quality — who beats whom?

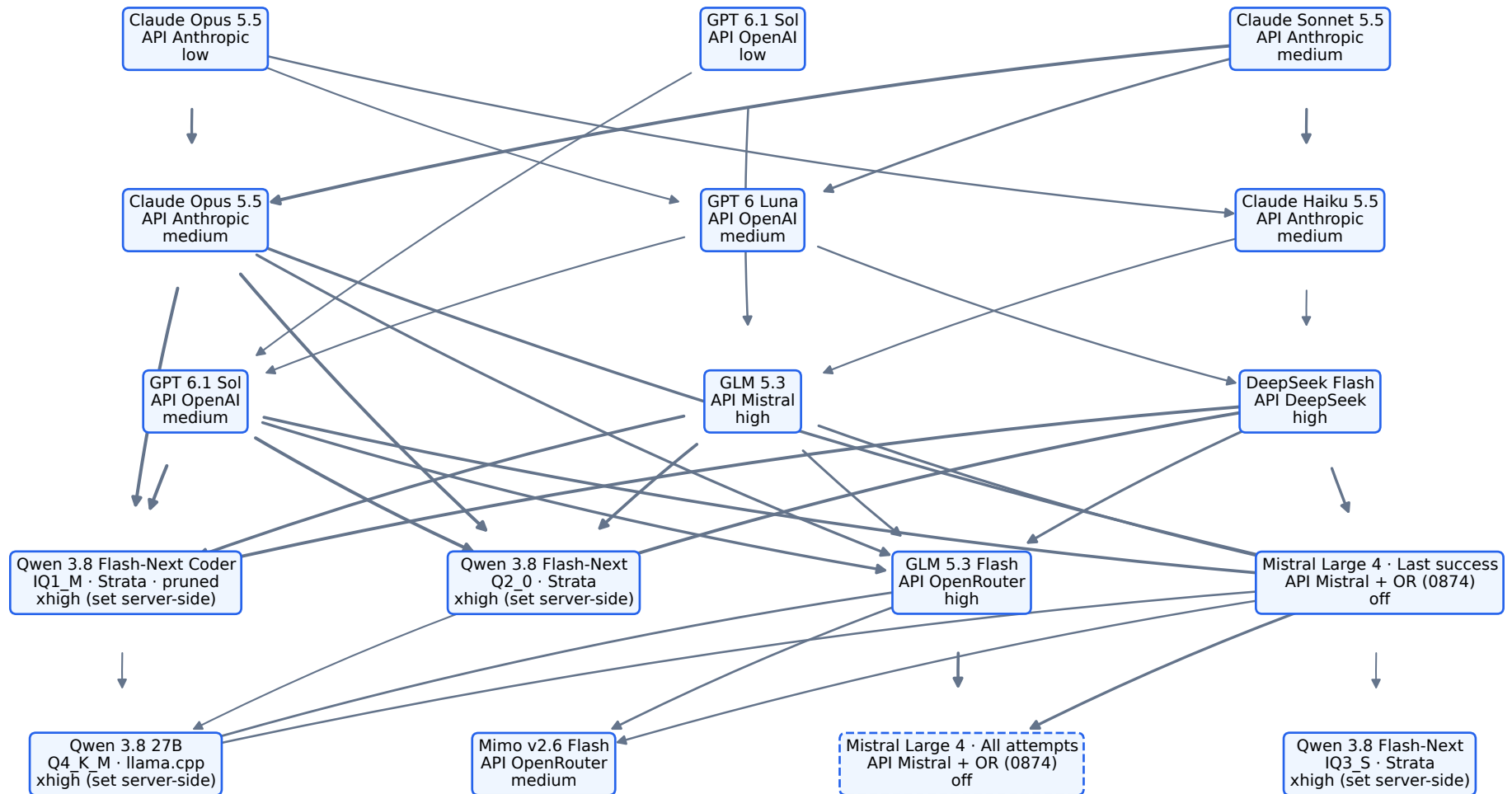
Accurate at the top · arrow towards Inaccurate · exact paired Wilcoxon · two-sided $p < 0.05$



These arrows are significant pairwise; ten tickets cannot guarantee the entire ranking at 95%.
Arrow thickness indicates effect intensity. No arrow ≠ equivalence.
An indirect path is not another significant test.

Speed — who beats whom?

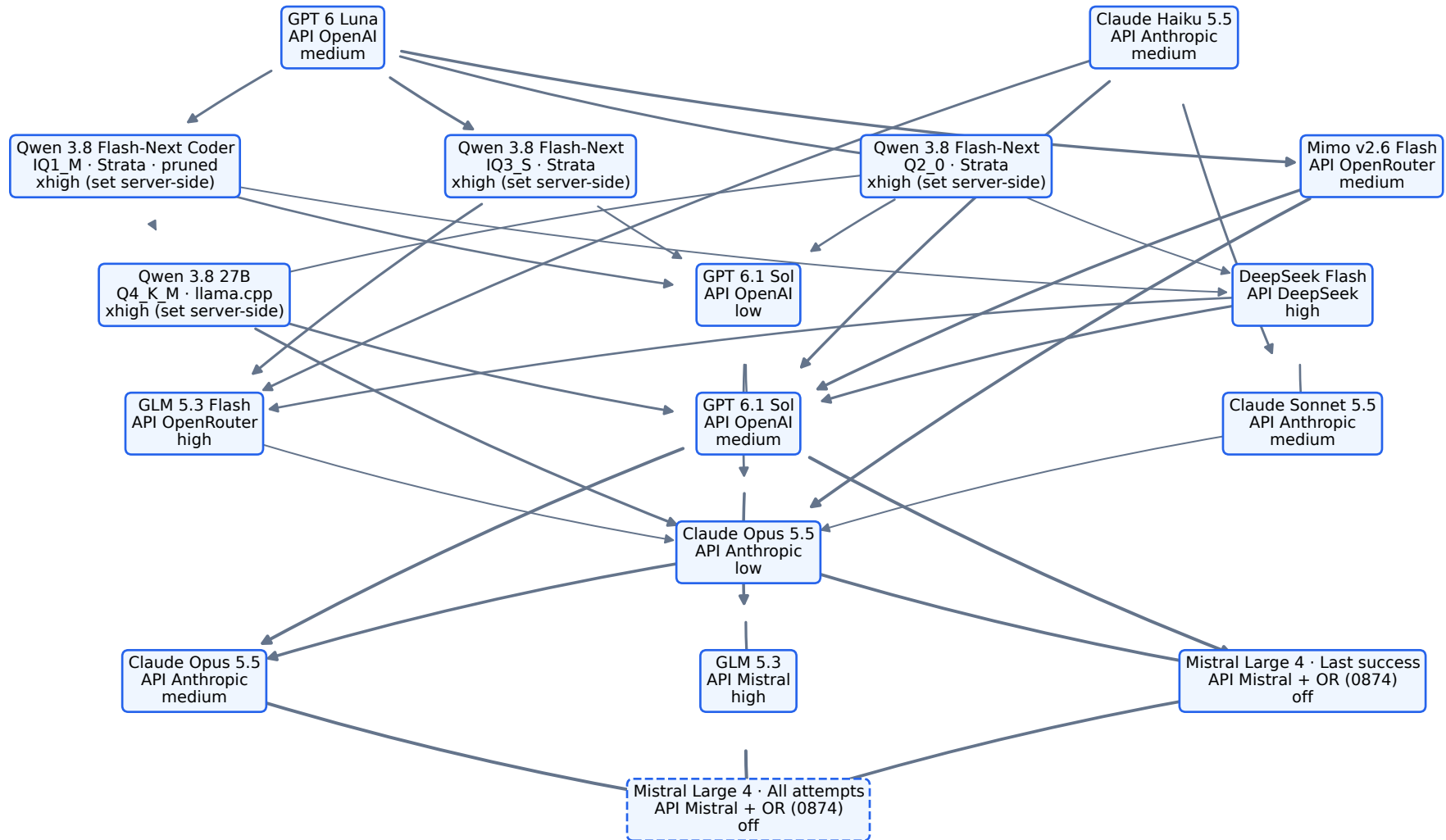
Fast at the top · arrow towards Slow · exact paired Wilcoxon · two-sided $p < 0.05$



These arrows are significant pairwise; ten tickets cannot guarantee the entire ranking at 95%.
Arrow thickness indicates effect intensity. No arrow $\neq$ equivalence.
An indirect path is not another significant test.

Cost — who beats whom?

Affordable at the top · arrow towards Expensive · exact paired Wilcoxon · two-sided $p < 0.05$



These arrows are significant pairwise; ten tickets cannot guarantee the entire ranking at 95%.
Arrow thickness indicates effect intensity. No arrow $\neq$ equivalence.
An indirect path is not another significant test.

Method and scope

Random sample of ten tickets from the author's project history. Each model replays the same tasks from their pre-resolution state, with the project instructions and a two-hour limit per attempt.

Pi provides the agent runtime: model access, tool calls and session management. Tickets come from projects using Imperial Dragon Harness (IDH). The prompt requires the historical instructions of each repository (AGENTS.md, tickets/AGENTS.md and associated rules). To evaluate each model separately, the tournament disables external reviews and the Gaze, review-pr and verify-gate panels: each model performs the task, runs mechanical checks and reviews its own work. The three judges intervene only after completion. This evaluates Pi with project instructions, rather than the full IDH workflow.

Boxing-style scoring out of 30: three judges start at ten and deduct points for faults according to severity: -3 , -2 , -1 or -0.5 . Judges inspect the work without knowing which model produced it. No usable result scores zero; a correctly explained inability to complete is judged normally.

The head judge is Gemini 3.1 Pro Preview; the other two are Grok 4.3 and MiniMax M2.5. Retained first-cycle results were judged by Gemini 3.1 Pro Preview, DeepSeek V4 Pro and Kimi K3. Scores are summed without judge weighting.

Local models run on a Lenovo ThinkStation P620: Ryzen Threadripper PRO 3945WX, 128 GB RAM, RTX A4000 + RTX 3060 GPUs (28 GB VRAM total). Local cost estimates electricity at 600 W and EUR 0.23/kWh, without hardware depreciation. API costs cover consumed tokens; the common conversion is USD 1.08 per EUR 1.

Scatterplots show mean quality, elapsed time and cost per ticket. Each ticket represents one average erg (unit of completed work). Delay is valued per erg per hour, in USD/(erg·h) or EUR/(erg·h), rather than as an hourly wage. The relevant cost is that of the delivered result: retries consume resources and delay delivery. Mistral last success retains the latest successful result per ticket. Mistral all attempts sums the costs and times of 25 retained direct attempts and the OpenRouter success on 0874, including interruptions, and retains the best score. Eight other OpenRouter attempts and one direct configuration-invalidated attempt are excluded. Costs are allocated by billed tokens; newer attempts remain estimates. Pareto eligibility requires complete series and mean quality of at least 15/30.

A May 2026 benchmark of Vietnam's thermal power inventory, with five repetitions per condition, showed the value of source documents and commercial models' repeatability. Here we assess work delivered with tools rather than the factual accuracy of an inventory.

- Pi : github.com/earendil-works/pi
- IDH : github.com/MinhHaDuong/ImperialDragonHarness
- Comparison data and scripts: frozen beta 0
- May 2026 benchmark: Beyond RAG (slides)

Method: comparisons and limitations

In the three directed acyclic graphs, arrows compare paired ticket results with a two-sided Wilcoxon signed-rank test at the 5% level. They point towards less accurate, slower or more expensive results. Transitively redundant arrows are removed for readability. No arrow does not establish equivalence; an indirect path is not an additional test.

The 5% threshold applies to each comparison, without a simultaneous guarantee for the whole graph. Holm correction controls the false-positive risk from multiple comparisons. With ten tickets and 136 comparisons per graph, the corrected threshold is unattainable for this test, even when a model wins all ten tickets. No arrows after correction therefore does not mean the models are equivalent: this sample cannot establish differences with that global guarantee. Mistral all attempts has a dashed outline in the DAGs. Qwen 3, Qwen 3.6 and GPT-6 Sol appear only in scatterplots.

Limitation: a single model performs all roles: orchestration, implementation and review of its own output. This measures its ability to complete a task end to end, rather than the performance of a harness routing roles to different models and assigning reviews to other models to reduce correlated errors. This design may disadvantage a worker model such as Haiku 5.5, or any model stronger in one role than in the others; the test does not measure that. The three external judges evaluate the final result without taking part in it.

Limitation: on ticket 0874, the last direct Mistral attempt reached the two-hour cap despite 195 successful API responses. The model spent a long time exploring the project and repeatedly testing, with missing modules and a missing hook in the environment. The identical-command repetition breaker does not catch this drift across varying commands. The successful OpenRouter result replaces this ticket in the last-success series; Mistral all attempts retains failed direct-attempt costs and times and adds the replacement. Mistral therefore combines two hosting routes. OpenRouter cost is the recorded cost of that run; direct costs follow the Mistral invoice calibration.

Limitation: ordinary series retain results after retries, whereas Mistral all attempts aggregates attempts. Repeatability means low dispersion across repetitions; first-attempt completion is another form of reliability. Heterogeneous retries prevent a uniform comparison of the latter here. “Attempted” counts retained attempts, including those invalidated by infrastructure or an incorrect starting state; these incidents are not all attributable to the model. Both Mistral rows count both providers, while all-attempts costs follow the documented scope. Mean generated tokens follow the runs used in each series, including reasoning when reported as output tokens; tokenizers differ across providers.

b2 retry: archives contain eleven attempts for ten tickets, including a 0333 attempt interrupted by SIGTERM after 17 min 24 s (exit code 143). The author confirmed requesting this pause to free the GPU for other work; the interruption was not based on output quality. The retained replay scored 30/30; the interrupted attempt had no score allowing best-score selection. This replacement and its excluded resource use mean the series is not ten first-attempt successes. Excluding 0333 from every series gives Qwen b2 and both Mistral series the same mean quality, 25.78/30 over nine tickets: b2 no longer has the sole highest mean. “Attempted” counts retained traces, not a guaranteed exhaustive history.

Candidates were selected ad hoc, without a claim to exhaustiveness. Which models, variants or runtimes would you like to see in the next comparison? Suggestions and feedback are welcome at minh.ha-duong@cnr.fr.